

HOMEVAULT

Product & Design Brief

A mobile home-management app for appliances, maintenance, repairs, warranties, documents, and household records

Product thesis

HomeVault gives homeowners one reliable place to answer: What do I own, what does it need, when was it serviced, what did it cost, and where is the documentation?

Prepared as a corrected replacement for the prior app brief

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1. Executive summary

HomeVault is a mobile-first household record and maintenance app. It helps homeowners and families capture appliances, systems, warranties, receipts, manuals, repairs, contractors, and recurring maintenance in one place.

The app is designed around infrequent but high-stress moments: a dishwasher leaks, an HVAC filter is overdue, a warranty is about to expire, a contractor asks for the model number, or a homeowner needs proof of purchase. Today, this information is usually scattered across email, paper folders, photos, spreadsheets, retailer accounts, and memory.

Core promise

Photograph it once. Find it when the house needs it.

Recommended launch shape

- Cross-platform mobile app using React Native and Expo.
- Offline-first local database with optional cloud backup and household sharing.
- MVP focused on asset records, documents, reminders, repairs, and exports.
- Photo-based model/serial capture using OCR, with human confirmation before saving.
- Freemium or low-cost lifetime purchase; avoid ads in the core product.
- No marketplace, contractor bidding, or broad home-automation integrations in the MVP.

2. Problem and opportunity

The homeowner problem

- Model and serial numbers are difficult to read and are often inaccessible after installation.
- Receipts and warranties are lost before they are needed.
- Home maintenance occurs on multiple schedules and is easy to forget.
- Repair history is fragmented across emails, invoices, calls, and memory.
- Replacement filters, bulbs, batteries, and parts are repeatedly researched.
- Household knowledge often lives with one person and is not shared.
- Home-sale preparation requires reconstructing years of upgrades and maintenance.
- Insurance claims require documentation that many households cannot quickly produce.

Why a dedicated app can win

The opportunity is not to become a generic to-do app. HomeVault should own the durable record of the physical home. The data becomes more valuable over time because each maintenance event, repair, purchase, and document adds to a persistent property history.

Current workaround	Weakness	HomeVault advantage
Camera roll	Hard to search; no structured fields or reminders	Photos linked to assets, rooms, documents, and events
Email search	Receipts and invoices are inconsistent and scattered	Central document vault with asset relationships
Spreadsheets	Manual entry, poor mobile use, weak photo support	Fast mobile capture and automatic calculations
Calendar reminders	Reminder has no model, part, manual, or history attached	Reminder opens the exact home record and task instructions
Retailer accounts	Only covers purchases from one retailer	Cross-retailer and cross-contractor home history
Paper binder	Not searchable, shareable, or backed up	Searchable household record with exports

3. Target users and use cases

User segment	Primary need	Key value
New homeowners	Understand unfamiliar systems and establish routines	Guided home setup and maintenance baseline
Busy families	Share home knowledge and avoid	Shared records, reminders, and

	missed tasks	assignments
DIY homeowners	Track parts, manuals, and repair history	Detailed asset records and service timeline
Older homeowners	Simplify home records and prepare for family support	Accessible organization and household sharing
Landlords with a few properties	Track units, appliances, and maintenance	Property-level organization without enterprise complexity
Home sellers	Document upgrades and maintenance	Exportable property history and improvement summary

Primary jobs to be done

19. When something breaks, help me identify the appliance or system and retrieve the right manual, warranty, part, and service history.
20. When maintenance is due, remind me and show exactly what to do and what supplies are required.
21. When I buy or install something, let me capture it quickly without filling out a long form.
22. When another household member needs information, let them find it without asking me.
23. When I sell, insure, or assess the home, produce a useful record of assets, repairs, and improvements.

4. Product principles and positioning

Principle	Design implication
Capture first, organize second	Let users save a photo or document immediately, then confirm extracted details.
The home is the organizing object	Records are grouped by property, room, asset, system, project, and event.
Human-confirmed automation	OCR and AI may suggest fields, but the user confirms important identifiers and dates.
Useful during stress	Critical information must be searchable and available offline.
Low administrative burden	The app should reduce household paperwork, not create another chore.
Durable ownership	Users can export their data and documents in common formats.
Privacy by default	Home data is sensitive and should not be monetized through ad targeting.

Positioning

Positioning statement

For homeowners who have household information scattered across photos, email, paperwork, and memory, HomeVault is a private home record and maintenance app that connects every appliance, document, task, repair, and part to the home it belongs to.

5. MVP scope

MVP capabilities

Capability	MVP behavior	Priority
Property setup	Create one home with address label, home type, purchase year, and optional photo	Must
Rooms and areas	Organize assets by room or area such as kitchen, garage, exterior, and utility room	Must
Asset records	Store category, brand, model, serial, purchase/install date, cost, photos, notes, and status	Must
Photo label capture	Photograph a product label and suggest brand/model/serial fields	Must
Documents	Attach receipts, warranties, manuals, invoices, and inspection reports	Must
Maintenance tasks	Create recurring or one-time tasks linked to an asset, room, or property	Must
Reminders	Local notifications with due, snooze, complete, and reschedule actions	Must
Repair history	Log issue, diagnosis, provider, date, cost, notes, and documents	Must
Parts and supplies	Save filter sizes, part numbers, battery types, paint colors, and replacement links	Should
Search	Search assets, model numbers, serial numbers, rooms, documents, and notes	Must
Household sharing	Invite one or more household members with owner/editor/viewer	Should

	roles	
Export	Export structured CSV/JSON and a printable property summary	Should
Cloud backup	Optional account-based encrypted sync and restore	Should

Explicitly out of scope for MVP

- Real-time retail pricing or automated shopping recommendations.
- Contractor marketplace or lead-generation system.
- Full accounting, tax, or depreciation functionality.
- Insurance-policy interpretation or claims advice.
- Smart-home device control.
- Automatic manufacturer warranty registration.
- Broad integrations with every retailer, utility, and service provider.
- Computer vision that claims to identify every appliance from a general photo.
- Automatic repair diagnosis presented as authoritative.

6. Core user workflows

A. Add an appliance or home system

24. Tap Add and choose Appliance/System.
25. Take a photo of the manufacturer label or upload an existing photo.
26. OCR suggests brand, model, serial number, and manufacturing date where present.
27. User confirms or corrects each extracted field.
28. Assign the item to a room or area.
29. Optionally attach receipt, warranty, manual, purchase date, and cost.
30. App suggests common maintenance tasks based on category, subject to user confirmation.

B. Complete maintenance

31. User receives a reminder such as Replace HVAC filter.
32. Task view shows the linked system, filter size, last completion date, notes, and any instructions.
33. User marks the task complete, records cost or a photo if useful, and optionally updates the next due date.
34. Completion is added to the asset and property history.

C. Handle a repair

35. Search or scan the affected asset.
36. Open model, serial, warranty, manual, and prior repair history.
37. Create a repair event and attach photos or an estimate.
38. Record contractor, cost, diagnosis, parts used, and resolution.

39. Create a follow-up task if additional work or inspection is required.

D. Prepare for a home sale or insurance claim

- 40. Choose Export property history.
- 41. Select assets, improvements, maintenance events, receipts, and repair records.
- 42. Generate a printable PDF summary and a structured data export.
- 43. Share the export without granting access to the entire account.

7. Information architecture and screens

Primary tab	Purpose	Key elements
Home	At-a-glance status	Upcoming tasks, overdue items, recent activity, quick add, home health summary
Inventory	Browse physical assets	Rooms, systems, appliances, supplies, search, filters
Maintenance	Manage recurring work	Due, upcoming, completed, seasonal, assigned tasks
Documents	Find household paperwork	Receipts, warranties, manuals, invoices, reports, document search
Profile/Household	Manage app and sharing	Homes, members, notifications, export, backup, subscription

Core screens

- Onboarding and create-home flow.
- Home dashboard.
- Room and area browser.
- Asset list and asset detail.
- Add asset with camera/OCR.
- Document viewer and document metadata editor.
- Maintenance task detail and completion flow.
- Repair/service event entry.
- Parts and supplies list.
- Search results.
- Household member and permissions screen.
- Export builder.

8. Data model

Entity	Important fields	Relationships
User	id, name, email, locale, timezone	Memberships, activity
Household	id, name, subscription, owner_id	Members, properties
Property	id, label, address, type, year_built, purchase_date	Rooms, assets, tasks, documents, projects
Room/Area	id, property_id, name, type, floor	Assets, documents, photos
Asset	id, room_id, category, brand, model, serial, install_date, purchase_date, cost, status	Documents, tasks, repairs, parts
Document	id, type, file_path, date, vendor, amount, OCR_text	Property, asset, task, repair, project
MaintenanceTemplate	category, title, interval, seasonal_rule, instructions	Suggested tasks
MaintenanceTask	id, scope, title, due_date, recurrence, assignee, status	Asset/room/property, completions
TaskCompletion	id, task_id, completed_at, cost, notes, photo	Task, user
RepairEvent	id, asset_id, issue, provider, diagnosis, resolution, cost, date	Documents, parts
Part/Supply	id, asset_id, name, part_number, size, quantity, link	Asset, maintenance task
Project/Improvement	id, property_id, title, dates, contractor, cost, notes	Documents, rooms, assets
HouseholdMember	household_id, user_id, role	Permissions

Important modeling decisions

- Documents should be attachable to multiple record types instead of living in a separate silo.
- Maintenance templates and actual household tasks must be separate; users own and can edit their schedule.
- Repairs and routine maintenance are different events but both appear on the property timeline.
- Assets should support non-appliance categories such as roof, paint color, flooring, windows, water shutoff, and breaker panel.
- Deleted records should use soft deletion while sync is enabled so household devices remain consistent.

9. Smart capture, OCR, and AI boundaries

Label and document capture

The highest-value automation is extracting structured information from photos and documents. The app should treat OCR as a suggestion system, not a source of truth.

Input	Suggested extraction	Required confirmation
Appliance label	Brand, model, serial, manufacture date, electrical rating	Model and serial
Receipt	Merchant, date, total, item description	Date, total, linked asset
Warranty	Coverage dates, provider, registration reference	Start/end dates
Invoice	Contractor, service date, total, work description	Provider, date, cost
Filter package	Dimensions, model, quantity	Size and linked system
Paint can label	Brand, line, color name/code, finish	Color code and room

AI boundaries

- Never overwrite user-entered values without confirmation.
- Display confidence and highlight uncertain characters in model and serial numbers.
- Do not present maintenance intervals as universally correct; identify them as suggested defaults.
- Do not diagnose electrical, gas, structural, or safety-critical problems as definitive.
- Do not upload documents to third-party AI services without explicit disclosure and consent.
- Support an OCR-only mode that does not use generative AI.

10. Reminders and maintenance engine

Task types

- Interval-based: every 30 days, 6 months, or 2 years.
- Seasonal: spring, summer, fall, or winter windows.
- Date-based: warranty expiration, inspection renewal, or planned service date.
- Usage-based: manually recorded hours, cycles, or quantity where useful.
- Condition-based: created after a repair or inspection, not automatically inferred.

Reminder behavior

State	Behavior
Upcoming	Visible on dashboard within a configurable look-ahead period
Due	Notification and prominent dashboard card

Overdue	Remains visible without sending excessive repeated alerts
Snoozed	New explicit date with reason retained in task history
Completed	Completion event recorded and next recurrence calculated
Skipped	Skipped occurrence recorded without falsely marking work complete

Reminder principle

A reminder must open into context: the exact asset, part number, instructions, last completion, and related documents.

11. Monetization

Recommended model

A privacy-oriented home record app should avoid advertising. The best initial model is a generous free tier with a paid household plan and an early lifetime option.

Plan	Illustrative scope	Illustrative price
Free	1 home, 20 assets, basic reminders, limited documents, local storage	\$0
HomeVault Plus	Unlimited assets/documents, cloud backup, household sharing, exports, advanced reminders	\$29.99/year or \$3.99/month
Early lifetime	Permanent Plus access for early adopters, subject to fair-use storage limits	\$49-\$79 one time
Multi-property	Several properties, expanded exports and member roles	\$69.99/year

Revenue expansion after validation

- Optional premium document storage tiers.
- Paid property-history report templates.
- Family/estate handoff package.
- Small-landlord multi-property plan.
- Privacy-respecting affiliate links for exact replacement parts only after user requests them.

12. Technical architecture

Recommended stack

Layer	Recommendation	Reason
Mobile	React Native + Expo + TypeScript	Fast cross-platform development and strong camera/document ecosystem
Local storage	SQLite with a typed data-access layer	Offline-first reliability and searchable structured records
Backend	Supabase/Postgres or equivalent	Authentication, relational sync, storage, row-level access controls
File storage	Encrypted object storage with signed URLs	Documents and photos are sensitive and potentially large
Subscriptions	RevenueCat	Cross-platform purchase and entitlement management
Notifications	Expo Notifications plus local scheduling	MVP reminders should work without server dependency
OCR	On-device OCR where practical; server OCR as opt-in fallback	Privacy, speed, and cost control
Analytics	Privacy-conscious product analytics with no document contents	Measure workflows without collecting sensitive data
Crash reporting	Sentry or equivalent	Operational visibility without inspecting user records

Offline-first strategy

- All core records remain usable without connectivity.
- New photos and documents queue for upload when cloud backup is enabled.
- Every record includes `created_at`, `updated_at`, `deleted_at`, `device_id`, and sync version metadata.
- Conflicts should favor explicit user choice for high-value fields such as model, serial, dates, and costs.
- Local reminders must not depend on a server job.

Suggested repository structure

apps/mobile, packages/domain, packages/database, packages/ui, packages/ocr, packages/sync, packages/export, supabase/migrations, and docs/product.

13. Privacy, security, and reliability

HomeVault will contain addresses, appliance identifiers, receipts, contractor information, photos, and potentially insurance-related records. This makes privacy a core product requirement rather than an optional enhancement.

- Encrypt data in transit and at rest.
- Use least-privilege household roles: owner, editor, and viewer.
- Use short-lived signed URLs for private documents.
- Do not expose storage object paths publicly.
- Require reauthentication for exports, household ownership transfer, and account deletion.
- Provide clear account and data deletion controls.
- Exclude OCR text and document contents from analytics and crash logs.
- Support device-level biometric lock as a premium or early feature.
- Back up relational data and object metadata separately from user files.
- Test restore workflows, not only backup creation.

Reliability requirements

- Asset and maintenance records must never disappear because a document upload failed.
- Users must be able to create and complete tasks offline.
- Exports must include a manifest so files can be matched to records.
- Notification rescheduling must be idempotent to prevent duplicate alerts.
- Migrations must preserve user-entered identifiers exactly, including leading zeros.

14. Analytics and success metrics

Metric	Why it matters	Early target
Activation	User creates home and saves first useful record	60% of installs
Capture completion	Label photo results in confirmed asset	70% of capture starts
Time to first asset	Measures onboarding friction	Under 3 minutes
Week-4 retained households	Indicates durable household value	25%+ early beta
Assets per active home	Shows record depth	10+ within 30 days
Reminder completion rate	Tests maintenance usefulness	40%+ of due tasks
Document attachment rate	Measures vault value	30%+ of assets
Household invite rate	Tests shared-knowledge value	15%+ of active homes
Export usage	Tests property-history value	Track qualitatively in beta
Paid conversion	Tests monetization	3-8% after product-market fit signals

Events to track

- home_created, room_created, asset_capture_started, ocr_completed, asset_saved
- document_added, reminder_created, reminder_snoozed, task_completed
- repair_logged, part_saved, household_invite_sent, export_generated
- subscription_started and subscription_cancelled

Do not record addresses, serial numbers, OCR text, document filenames, notes, or contractor names in analytics.

15. Validation plan

Before full implementation

- 44. Create a clickable prototype for Home, Inventory, Add Asset, Asset Detail, and Maintenance.
- 45. Test with 12-20 homeowners across new homeowners, families, DIY users, and older homeowners.
- 46. Ask each person to add a real appliance from a label photo and retrieve it later.
- 47. Observe whether they understand the relationship between an asset, document, task, and repair.
- 48. Ask participants to bring one real household record they currently struggle to manage.
- 49. Offer a paid early-access reservation rather than relying only on stated interest.

Beta test

- Recruit 30-50 households.
- Require each beta home to add at least five real assets.
- Run for six to eight weeks so at least one maintenance cycle occurs.
- Interview users after their first real repair, reminder, or document retrieval.
- Measure setup abandonment and which fields users consistently skip.

Go/no-go evidence

Signal	Continue	Rework
Capture	Most users can save an asset in under 3 minutes	OCR correction or forms create high abandonment
Return value	Users reopen records during maintenance or repair	App is used once as a static inventory
Retention	Meaningful week-4 household retention	No reason to reopen after setup
Payment	Users accept annual or lifetime pricing	Users only value it as a free utility
Sharing	Some households invite a second member	Knowledge remains single-user and non-collaborative

16. Roadmap

Phase	Objective	Scope
0 - Validation	Prove workflow and willingness to pay	Prototype, interviews, landing page, paid beta
1 - Local MVP	Deliver core value without backend dependency	One home, rooms, assets, OCR capture, documents, tasks, local notifications, search
2 - Cloud household	Make records durable and shared	Authentication, backup, sync, household roles, file storage
3 - Property history	Increase long-term value	Repairs, improvements, advanced exports, contractor records, parts
4 - Smart assistance	Reduce repetitive entry	Document extraction, maintenance suggestions, duplicate detection, manual lookup links
5 - Multi-property	Expand addressable market	Landlord plan, property switching, portfolio exports, role refinements

17. Risks and mitigations

Risk	Impact	Mitigation
Users resist setup effort	Low activation	Photo-first capture, progressive forms, import later
OCR makes identifier mistakes	Wrong parts or warranty lookup	Confidence indicators and mandatory confirmation
Maintenance suggestions feel generic	Low trust	Make all suggestions editable and cite manufacturer guidance when available
Cloud storage costs grow	Weak unit economics	Compression, quotas, storage tiers, local-first defaults
App is opened too infrequently	Low retention	Useful reminders, repair workflow, document retrieval, property timeline
Scope expands into contractor marketplace	Complexity and conflicts	Keep provider records private and user-entered
Sensitive home data is exposed	Severe trust damage	Private-by-default architecture, access controls, audits, export/delete controls

Users confuse the app with professional advice	Safety and liability risk	Clear educational framing and safety escalation language
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18. Initial implementation backlog

Epic	Initial stories
Foundation	Expo project, TypeScript rules, navigation, design tokens, local database, migrations, error handling
Onboarding	Create household, create home, suggested rooms, notification permission
Inventory	Asset list, asset detail, create/edit asset, room assignment, photos
Capture	Camera flow, crop/rotate, OCR extraction, field confirmation, confidence display
Documents	Import file/photo, document type, link to asset, viewer, delete/export
Maintenance	Task templates, create recurrence, local scheduling, due list, completion history
Repairs	Create repair event, cost/provider fields, documents, timeline display
Search	Full-text search over structured fields and OCR text, filters by room/category/type
Export	CSV/JSON export, PDF property summary, ZIP file manifest
Sync	Authentication, row-level security, household memberships, background sync, conflict handling
Monetization	Entitlements, free limits, paywall, restore purchases
Quality	Unit tests, database migration tests, offline tests, notification tests, accessibility checks

First shippable release

Release definition

A homeowner can create a home, photograph an appliance label, confirm model and serial information, attach a receipt or manual, create a recurring maintenance task, receive a reminder, complete the task, and later retrieve the complete record offline.